

Lakshyaraj Singh Rao

Full-Stack Engineer · AI Systems · Backend · DevOps

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SUMMARY

Backend-focused engineer who ships production AI products end-to-end. Solo-built and operates Homesty.ai — a live, commission-generating real-estate AI platform — with a 5-layer anti-fabrication system, 165 production deployments, and 0 critical Sentry classes firing under live traffic. Comfortable across the full stack: REST API design, Prisma ORM, RAG infrastructure, Docker, CI/CD, observability.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), C++, SQL

Backend: Node.js, Express.js, Next.js 15, REST APIs, JWT Auth, Prisma ORM, Zod

AI / LLM: GPT-4o, Claude Sonnet, Vercel AI SDK, RAG (pgvector), Streaming, Prompt Engineering

Databases: PostgreSQL (Neon), pgvector, MongoDB, Redis (Upstash), RabbitMQ

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, Vercel, AWS EC2, Kubernetes, ArgoCD

Observability: Sentry, Prometheus, Grafana, Winston, Playwright, Vitest

EXPERIENCE

Founding Engineer & Co-Founder — Homesty AI Technology LLP

Mumbai, India · Mar 2026 – Present · Entity incorporating May 2026 · homesty.ai

Solo engineer on the production stack. End-to-end ownership: system prompt design, RAG infrastructure, full-stack development, security, observability, incident response. Co-founded with Balvir Singh Rao (operations and legal).

- Shipped buyerchat to production: Next.js 15 + Postgres + pgvector + Prisma 7 + GPT-4o + Claude + Sentry + Upstash Redis + Docker. 16 RERA-verified projects indexed for South Bopal and Shela, Ahmedabad.
- 165 production deployments. 0 critical Sentry classes firing under live traffic. 159 unit tests passing. Live commission model.
- Engineered 5-layer anti-fabrication architecture: PART 0 hard-stop rules + onChunk markdown abort + 17-class regex audit + source-provenance API blockade + GUARD_LIST RAG grounding.
- Closed 8 production hallucination classes including OTP-simulation, fabricated entity names, and invented founding years.
- 8-stage chat pipeline: per-IP rate limit → intent classifier → RAG retrieval → GPT-4o streaming → post-stream audit → DB log + Sentry alert.
- Built 7-module Decision Engine: scores projects against buyer preferences, generates Strong Buy / Buy with Conditions / Avoid tags with honest concern disclosures.
- pgvector retriever with 600ms timeout + 0.30 cosine floor — silently drops weak matches instead of feeding garbage to the model.
- Security: prompt injection protection, PII stripping, origin checks, PRICING_LOCKED canonical surface, per-IP capture rate limiting.

PROJECTS

vercel-ai-eval

Drop-in eval framework for Vercel AI SDK projects. Golden-dataset, LLM-as-judge, and structural assertion runners with GitHub Actions PR integration. TypeScript, npm-publishable. Apache 2.0.

rag-starter

Production RAG template extracted from buyerchat. pgvector retrieval, embed-writer, cosine-floor, provenance API for grounded citations. Apache 2.0.

[guardrail-proxy](#)

Streaming LLM safety. Mid-chunk abort when audit catches violation — user never sees a bad response finish rendering. 17-class regex audit, PII scrub. Apache 2.0.

[devops-showcase](#)

Kubernetes + ArgoCD GitOps stack. Canary deploys, Prometheus, Grafana. Apache 2.0.

[k8s-deploy-tracker](#)

Go service that consumes Kubernetes deploy events and feeds failing ones to GPT-4 for one-paragraph summaries. Apache 2.0.

[stream-bench](#)

Benchmarks for LLM streaming: time-to-first-token, throughput, total latency across OpenAI, Anthropic, Bedrock. Apache 2.0.

[EDUCATION](#)

Manipal University Jaipur — B.Tech, Computer Science

Graduating 2026

[RECOGNITION](#)

- **Production Reliability:** 165 deployments shipped, 0 critical Sentry classes firing, 159 unit tests passing — homesty.ai live traffic.
- **Anti-Fabrication Architecture:** Closed 8 production hallucination classes via PART 0 hard rules + 17-class regex audit + GUARD_LIST RAG grounding.